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DAY HRS MIN SEC

Gridlock Hackathon 2.0

LIVE

May 26, 2026, 10:00 PM IST (Asia/Kolkata) - Jun 07, 2026, 11:59 PM IST (Asia/Kolkata)

Traffic demand prediction

Max. score: 100

Cities worldwide are increasingly turning to AI-powered solutions to tackle the issue of traffic congestion. This problem disrupts the smooth flow of transportation and poses a significant barrier to economic growth. To address this challenge effectively, the first step is to understand travel demand and patterns within urban areas comprehensively. By harnessing the power of AI, cities, and regions aim to gather critical insights into transportation dynamics. This will enable them to implement data-driven strategies and solutions to alleviate traffic congestion and promote more efficient mobility. Ultimately, this endeavor will foster economic development and prosperity.

Task

Design a system that helps us provide valuable insights into passenger travel patterns, booking behavior, and trip cancellations, which can be used for various analyses and predict demand in the travel industry.

Dataset description

The dataset folder contains the following files:

- **train.csv**: 77299 x 11
- **test.csv**: 41778 x 10
- **sample_submission.csv**: 5 x 2

Variable description

The columns provided in the dataset are as follows:

Column name	Description
Index	Represents the unique identification of datapoint
geohash	Represents geographic information regarding a place
day	Represents the day when the information is recorded
timestamp	Represents the timestamp of the record inserted in the system

RoadType	Represents the type of road in the nearby location
NumberofLanes	Represents the number of roads/lanes present in the location
LargeVehicles	Represents whether large vehicles are permitted on the specific roads/lanes
Landmarks	Represents whether there are any landmarks near the location
Temperature	Represents the temperature of the place
Weather	Represents the weather of the place
demand	Represents the demand of the traffic at the timestamp

Evaluation metric

```
score = max(0,100*(metrics.r2_score(actual,predicted)))
```

Result submission guidelines

- The index is "Index" and the target is the "demand" column.
- The submission file must be submitted in .csv format only.
- The size of this submission file must be 41778 x 2.
- Ensure that your submission file contains the following:
  - Correct index values as per the test file
  - Correct names of columns as provided in the *sample_submission.csv* file

Instructions

1. Click *Download dataset* to download the dataset.
2. Solve the problem in your local environment.
3. Save the predictions in a *.csv* file.
4. Click *Upload File* (under the *Upload File* section) to upload your prediction file (*.csv*).
5. Click *Upload File* (under the *Upload Source Code* section) to upload your *.ipynb* file along with any presentation file.
6. Add any instructions or comments in the *Your Answer* section.
7. Click *Submit*.

Download dataset

Upload Prediction File

Please upload the prediction file in the format as stated in the problem.

Choose File No file chosen

Submit & Evaluate

Upload Source Files

You need to submit a zip or tar archive consisting of a text file explaining your approach, details about feature engineering, tools you used and the relevant source files.

Choose File No file chosen

Upload